LAB 01 REPORT

Building the Invisible

Penetration Testing Lab Environment

Layered Anonymity Architecture Design & Implementation

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Course:

Intrusion Testing

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Section 1 — Introduction & Rules of Engagement

1.1 Introduction

This lab focuses on designing and implementing a fully isolated penetration testing environment using a layered anonymity architecture. The primary objective is to ensure that all penetration testing activities remain private, controlled, and legally compliant within defined boundaries.

The environment integrates the following core components:

- OPNsense — Firewall and router, enforcing network segmentation and traffic policy
- WireGuard — Encrypted VPN tunnels providing dual-layer identity separation
- Tor via Whonix — Final anonymization layer, ensuring exit-node IP concealment

This layered architecture collectively ensures:

- The operator's real IP address is never exposed to target systems
- Traffic is encrypted at multiple independent layers
- All testing activities remain within the defined scope and are legally compliant

1.2 Rules of Engagement (RoE)

Scope

- Network range: 192.168.10.0/24
- Whonix Gateway & Workstation VMs
- Kali Linux attacker machine
- Target systems: Metasploitable, DVWA, Juice Shop

Out of Scope

- Host (bare-metal) machine
- Any public internet address
- ISP network infrastructure

Authorized Techniques

- Network scanning (Nmap)
- VPN configuration and verification

- Traffic analysis (Wireshark)
- Web application testing (DVWA, Juice Shop)

Prohibited Actions

- Any attack directed at external or public systems
- Traffic generation outside the defined lab scope
- Denial-of-Service (DoS) attacks of any kind

Evidence Handling

- All screenshots stored locally on the lab workstation
- Final report submitted via the LMS portal

Section 2 — Network Architecture

2.1 Architecture Overview

The following traffic flow represents the layered anonymity chain implemented in this lab:

```
Kali / Whonix Workstation
↓  (Application traffic)
Whonix Gateway  (Tor network)
↓  (Tor-encrypted traffic)
OPNsense Firewall  (LAN: 192.168.10.1)
↓  (WireGuard tunnel 2)
VPN 2  (Identity separation layer)
↓  (WireGuard tunnel 1)
VPN 1  (ISP concealment layer)
↓
Internet
```

2.2 Architectural Rationale

This design follows a defense-in-depth anonymity model, where each layer independently prevents a specific category of information leakage:

- VPN 1 — Conceals testing traffic from the Internet Service Provider (ISP), preventing observation of destination addresses.
- VPN 2 — Separates operator identity from destination by introducing an additional routing hop unknown to the exit point.
- Tor (via Whonix) — Provides the final anonymization layer through onion routing, making traffic correlation extremely resource-intensive even for sophisticated adversaries.

The combination of these independent layers ensures that no single point of failure can de-anonymize the operator's identity during testing activities.

Section 3 — LLM-Assisted Research Process

Large Language Models (LLMs) were used as a supplementary research tool during the configuration phases of this lab. Each AI-generated response was independently verified against official documentation or live system behaviour before implementation.

3.1 Research Log

Problem 1	OPNsense LAN/WAN Configuration

- Prompt: How to configure LAN and WAN interfaces in OPNsense
- Result: Step-by-step interface assignment and IP configuration procedure
- Verification method: Applied directly in virtual machine and cross-referenced with OPNsense documentation
- Accuracy assessment: Correct

Problem 2	WireGuard VPN Setup in OPNsense

- Prompt: WireGuard tunnel configuration within OPNsense
- Result: Detailed setup steps including key generation, peer configuration, and interface assignment
- Verification method: Compared output with actual WireGuard configuration files and verified handshake
- Accuracy assessment: Correct

Problem 3	Tor Connection Verification

- Prompt: How to verify that traffic is correctly routed through Tor
- Result: Suggested the curl command: curl https://check.torproject.org/api/ip
- Verification method: Executed command in Whonix Workstation; confirmed Tor exit node IP
- Accuracy assessment: Correct

3.2 Analysis & Observations

LLM assistance proved effective for accelerating configuration lookups and troubleshooting common setup issues. However, a critical caveat applies in the cybersecurity context:

All AI-generated responses were independently verified before implementation. Blind trust in AI-generated answers is methodologically unacceptable in security work — configuration errors in a penetration testing environment can result in IP leaks, scope violations, or incomplete isolation.

Section 4 — Build Log

The lab environment was constructed in sequential phases to ensure each layer was fully verified before the next was introduced.

Phase 1 — Network Design

- Defined overall network topology and traffic flow
- Assigned IP address ranges and subnets
- Planned routing policy between layers

Phase 2 — OPNsense Installation & Base Configuration

- Created OPNsense VM: 2 GB RAM, 2 vCPUs, VirtualBox
  - WAN adapter: attached to NAT
  - LAN adapter: attached to Internal Network
- Assigned LAN IP address: 192.168.10.1/24
- Enabled DHCP server on the LAN interface

Phase 3 — OPNsense Web Interface Configuration

- Accessed the OPNsense web GUI at https://192.168.10.1
- Configured system hostname and domain
- Set DNS server to 9.9.9.9 (Quad9 — privacy-respecting resolver)
- Disabled IPv6 globally to prevent IPv6 leak vectors

Phase 4 — WireGuard VPN Tunnel Configuration

- Created two independent WireGuard tunnel instances (VPN1 and VPN2)
- Generated cryptographic key pairs for each tunnel endpoint
- Configured peer public keys and allowed IP ranges
- Assigned WireGuard interfaces within OPNsense
- Verified successful handshake for both tunnels via the OPNsense VPN status panel

Phase 4.5 — Routing Policy

- Configured policy routing: LAN traffic routed through VPN2
- VPN2 egress traffic routed through VPN1 to the internet

Phase 4.6 — Whonix Integration & Tor Verification

- Imported Whonix Gateway and Whonix Workstation OVA images into VirtualBox
- Configured Whonix Gateway network interfaces to route through OPNsense LAN
- Verified Tor connectivity using the following command on the Whonix Workstation:

```
curl https://check.torproject.org/api/ip
```

The command returned: {"IsTor": true, "IP": "193.189.100.200"} — confirming successful Tor routing.

Section 5 — Verification Results

A comprehensive suite of leak tests was performed to validate the integrity of the anonymity architecture. All tests were conducted from the Whonix Workstation after full configuration was applied.

Test	Method / Result	Verdict
IP Leak	ipleak.net — Tor exit node IP displayed	✓ PASS
DNS Leak	No ISP DNS servers visible in results	✓ PASS
WebRTC Leak	No public IP address exposed via WebRTC	✓ PASS
IPv6 Leak	IPv6 connection failed (intentionally disabled)	✓ PASS
Kill Switch	VPN disabled → traffic fully stopped	✓ PASS

All five verification tests passed without exception. The environment successfully prevents real IP exposure, DNS leakage, WebRTC-based fingerprinting, and uncontrolled traffic when the VPN connection drops.

Section 6 — Threat Model

The following threat model identifies the primary attack vectors relevant to this anonymity architecture, along with the corresponding mitigations applied.

Threat Vector	Risk Description	Mitigation Applied
DNS Leak	ISP can observe domain resolution queries	Unbound DNS resolver + OPNsense firewall rules
IPv6 Leak	IPv6 traffic can bypass VPN tunnel	IPv6 fully disabled on all interfaces
WebRTC Leak	Browser may expose real IP via WebRTC	Whonix hardened browser configuration
Kill Switch Failure	Real IP exposed if VPN drops unexpectedly	OPNsense firewall rules block non-VPN traffic
Traffic Correlation	Nation-state adversary correlates entry/exit nodes	Tor layered over double VPN chain

6.1 Advanced Adversary Consideration

A nation-state level adversary would most likely attempt traffic correlation attacks — monitoring both the entry point to VPN1 and the Tor exit node simultaneously to infer the operator's identity through timing analysis.

However, the double VPN chain combined with Tor's multi-hop onion routing significantly increases the resource and coordination requirements for such an attack. The probability of successful de-anonymization is substantially reduced when multiple independent anonymization layers are traversed.

Section 7 — Conclusion

This lab successfully demonstrated the design, implementation, and verification of a fully isolated penetration testing environment with layered anonymity. The architecture integrates OPNsense firewall management, WireGuard VPN chaining, and Tor-based traffic anonymization through Whonix.

All five leak verification tests passed, confirming that:

- No real IP address was exposed through any tested channel
- DNS queries were fully contained within the defined resolver chain
- WebRTC and IPv6 leak vectors were fully neutralized
- Kill switch behaviour correctly halted all traffic upon VPN disconnection

Key Takeaway

Security is not about tools — it is about correct configuration.

The combination of OPNsense, dual WireGuard VPN tunnels, and Tor provides a robust foundation for secure and anonymous penetration testing operations. This architecture can be extended in future labs to include traffic analysis countermeasures, network intrusion detection systems, and additional target environment isolation.